

Question

S crystal contains three types of ions: tetrahedral **A**, **B**, and heavy atom coordination polyhedron octahedral **C**, in a ratio of 1 : 1 : 1. It belongs to the orthorhombic crystal system, with cell parameters $a = 6.9550\text{\AA}$, $b = 6.1420\text{\AA}$, $c = 11.2180\text{\AA}$. There are two chemical formulas in one unit cell. Heating **S** to 100°C results in a weight loss of 44.05%, and further heating results in another weight loss of 10.61%, yielding a stable compound **X**. **X** contains 27.83% of the non-metallic element **Y**. Which of the following statements are correct:

1. All elements in **S** are macroelements in plants.
2. The sum of all atoms in the chemical formula of **X** is 12.
3. From the transformation of **S** to **X**, 14 moles of gas are produced for every 1 mole of **X** formed.
4. The density of the crystal is 1.701g/cm^3 .
5. The stoichiometric ratio of each species in the second step heating chemical equation is 2:1:2:1.

A. All other options are incorrect

B. 1.5.

C. 1.2.4.

D. 2.4.5.

E. 3.4.

F. 2.5.

G. 1.4.5.

H. 2.3.5.

Answer

Correct Answer: **G**

Detailed Explanation

Based on the weight loss from water release, the octahedral **C** should be $\text{M}(\text{H}_2\text{O})_6^{n+}$

$$(18.016 \times 6)/0.4405 \times (1 - 0.4405 - 0.1061) \times 0.2783 = 30.97 \text{ g/mol}$$

Therefore, **Y** is element *P*

CHECKPOINT

1 PTS

Therefore, **Y** is element *P*

Considering that the tetrahedral anion should be PO_4^{3-} , the other tetrahedral cation is most likely NH_4^+ (because there is no residual weight after heating)

CHECKPOINT

1 PTS

Tetrahedral anion should be PO_4^{3-}

CHECKPOINT

1 PTS

Tetrahedral cation is most likely NH_4^+

From the remaining mass, the metal ion is $\text{Mg}(\text{H}_2\text{O})_6^{2+}$

1 is correct.

CHECKPOINT

1 PTS

Metal ion is $\text{Mg}(\text{H}_2\text{O})_6^{2+}$

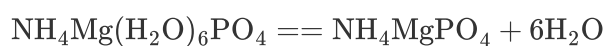
X should be pyrophosphate. Based on the mass fraction, it can be deduced that **X** is $\text{Mg}_2\text{P}_2\text{O}_7$, 2 is incorrect.

CHECKPOINT

1 PTS

X is $\text{Mg}_2\text{P}_2\text{O}_7$

The chemical equations involved in the decomposition:



CHECKPOINT

1 PTS



5 is correct

From **S** to **X**, for every 1 mole of **X** generated, 15 moles of gas will be produced, 3 is incorrect.

$$\rho = ZM/N_A V = (2 * 245.4)/(N_A * 6.955 * 6.142 * 11.218 * 10^{-24}) = 1.701 \text{g/cm}^3$$

CHECKPOINT

1 PTS

$$\rho = ZM/N_A V = (2 * 245.4)/(N_A * 6.955 * 6.142 * 11.218 * 10^{-24}) = 1.701 \text{g/cm}^3$$

4. is correct

Therefore, G is correct.